

SECR2043 OPERATING SYSTEMS

Name : _____
Student ID : _____
Section : _____

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	Output
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
4.	Change your current working directory to "os_lab".	
	Command	Output
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output

6.	Display the content of "hello.txt".	
	Command	Output
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
9.	Delete "hello.txt".	
	Command	Output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output

8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
3.	Run the executable file "hello" and display its output.	
	Command	Output
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output

8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	Total Mark:	